

Novelty and Related-Work Statement

Privacy, financial integrity, adaptive policy and national-scale qualification across independent sovereign ledgers

Research gap

CBDC projects have demonstrated important components, including two-tier ledgers, compliance proofs, privacy mechanisms, cross-border settlement and high-throughput experimental engines. Public evidence remains fragmented across different architectures, market structures and evidence standards.

Scoped novelty

The manuscript does not claim novelty for consensus, atomic commit, zero-knowledge proofs, differential privacy, graph analytics, sanctions matching, holding limits or queueing models individually. Its contribution is the explicit composition of these mechanisms around monetary accounting, sovereign-ledger isolation and a claims-to-evidence discipline.

Distinctive elements

- Cross-border terminal certificates bind unequal-currency FX quotes, bilateral compliance authorizations and final domestic prepare receipts.
- Privacy is evaluated through learned linkage attacks, while formal differential privacy is restricted to a registered release service.
- AML/CFT and sanctions results are translated into prospective outcomes and analyst workload rather than represented as automatic legal effectiveness.
- Economic limits are partially identified and adaptively governed rather than declared universal.
- National scale is approached through analytical lower bounds, a falsifying digital twin, physical acceptance criteria and pilot evidence.

Claim boundary

The manuscript supports a reproducible architecture and qualification methodology. It does not claim sovereign certification, live institutional effectiveness, an optimal global policy limit or production capacity based on local database measurements.

Editorial positioning

The paper is best positioned as interdisciplinary design-science and empirical systems research for a journal covering digital finance, payment systems, privacy engineering, financial regulation, central banking or critical financial infrastructure.